

DSA0601
DATA HANDLING AND VISUALIZATION
SET 5: WEBSITE ANALYTICS

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EXPERIMENT: WEBSITE ANALYTICS

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Objectives

- Visualize the daily trend in website page views over time.
- Identify the top days based on click-through rate.
- Display the distribution of user interactions using a stacked area chart.
- Prepare data for an interactive Tableau dashboard.
- Interpret the main traffic and engagement patterns in the dataset.

Program 1: Line Chart – Daily Page Views

A line chart is used to visualize how daily page views change over time.

```
# PROGRAM 1: LINE CHART - DAILY PAGE VIEWS
# -----
line_chart <- ggplot(
  website_traffic,
  aes(x = Date, y = Page_Views)
) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_point(color = "steelblue", size = 3) +
  labs(
    title = "Daily Website Page Views",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()

print(line_chart)

# -----
```

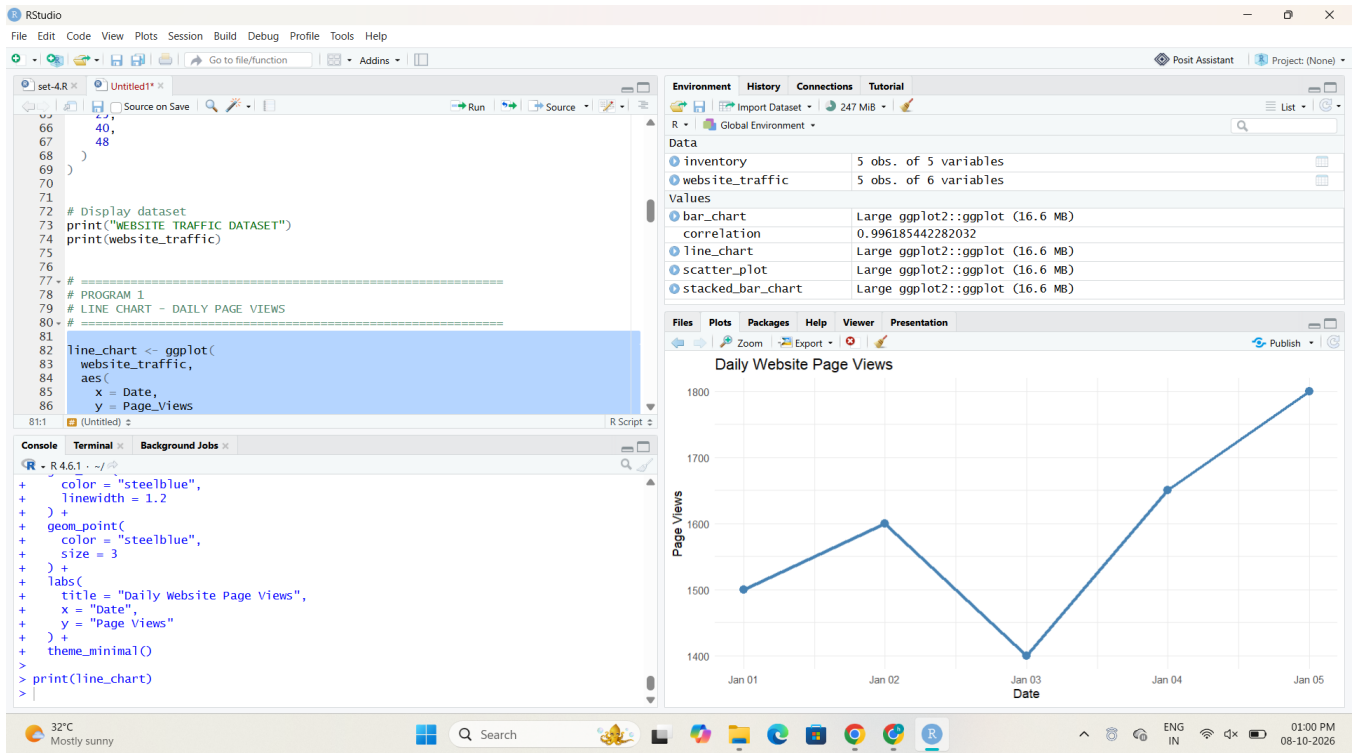


Figure 1: RStudio output – Daily Website Page Views

Observation: Page views increased from 1,500 on January 1 to 1,600 on January 2, decreased to 1,400 on January 3, and then increased to 1,650 and 1,800 on January 4 and January 5. The highest page-view count is 1,800 on January 5.

Program 2: Bar Chart – Top N Days by Click-Through Rate

The top-N bar chart ranks the supplied days according to click-through rate. Since five days are provided, N = 5 displays all five days.

```
# PROGRAM 2: BAR CHART - TOP N DAYS BY CTR
# -----
N <- 5

top_days <- website_traffic[
order(-website_traffic$Click_Through_Rate),
]

top_days <- head(top_days, N)

bar_chart <- ggplot(
top_days,
aes(x = reorder(as.character(Date), Click_Through_Rate),
y = Click_Through_Rate)
) +
geom_col(fill = "orange") +
geom_text(
aes(label = paste0(Click_Through_Rate, "%")),
vjust = -0.5
) +
labs(
title = paste("Top", N, "Days by Click-Through Rate"),
x = "Date",
y = "Click-Through Rate (%)"
) +
theme_minimal()

print(bar_chart)

# -----
```

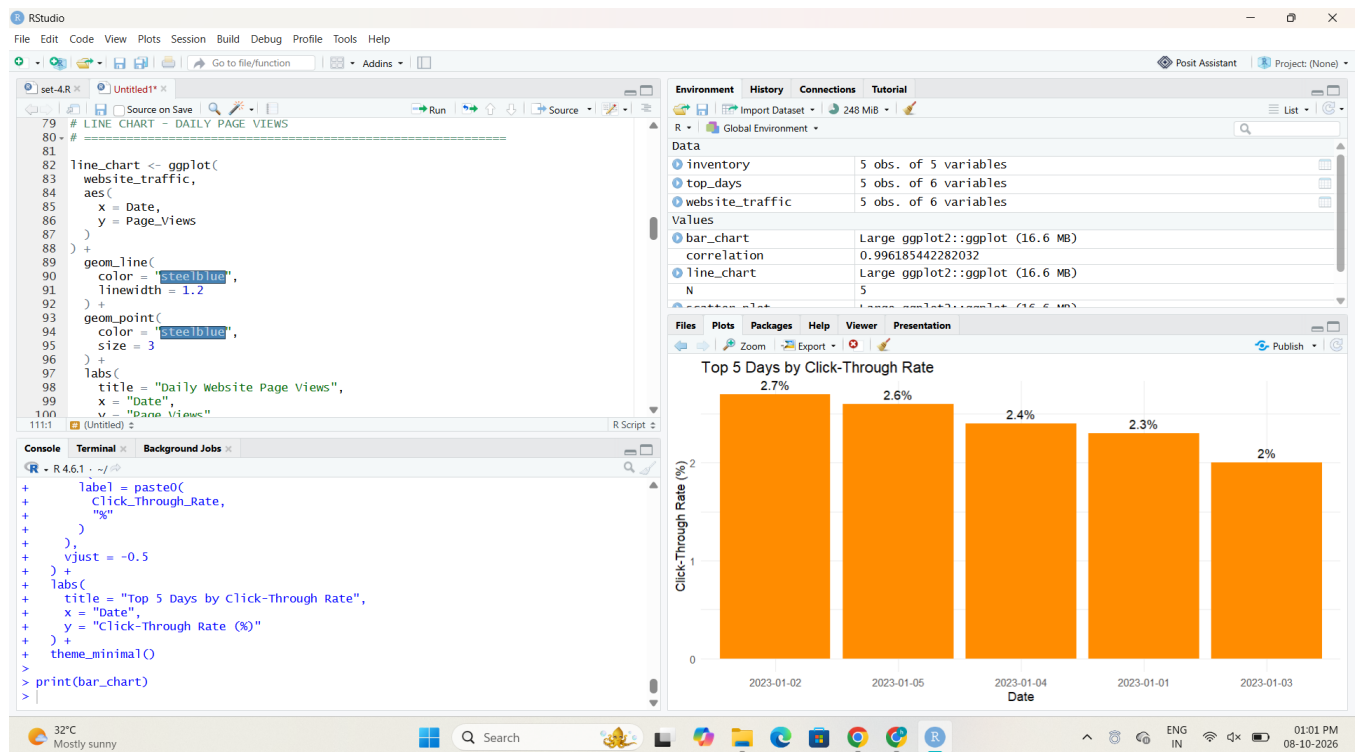


Figure 2: RStudio output – Top 5 Days by Click-Through Rate

Observation: January 2 has the highest click-through rate at 2.7%, followed by January 5 at 2.6%, January 4 at 2.4%, January 1 at 2.3%, and January 3 at 2.0%.

Program 3: Stacked Area Chart – User Interactions

The supplied Website Traffic dataset contains page views and click-through rate but does not provide likes, shares, or comments. Therefore, sample interaction values were added only to demonstrate the requested stacked area chart.

```
# PROGRAM 3: STACKED AREA CHART - USER INTERACTIONS
# -----
interaction_data <- data.frame(
  Date = rep(website_traffic$Date, times = 3),
  Interaction = rep(
    c("Likes", "Shares", "Comments"),
    each = nrow(website_traffic)
  ),
  Count = c(
    website_traffic$Likes,
    website_traffic$Shares,
    website_traffic$Comments
  )
)

stacked_area_chart <- ggplot(
  interaction_data,
  aes(x = Date, y = Count, fill = Interaction)
) +
  geom_area(alpha = 0.8) +
  labs(
    title = "Distribution of User Interactions",
    x = "Date",
    y = "Number of Interactions",
    fill = "Interaction Type"
  ) +
  theme_minimal()

print(stacked_area_chart)

# -----
```

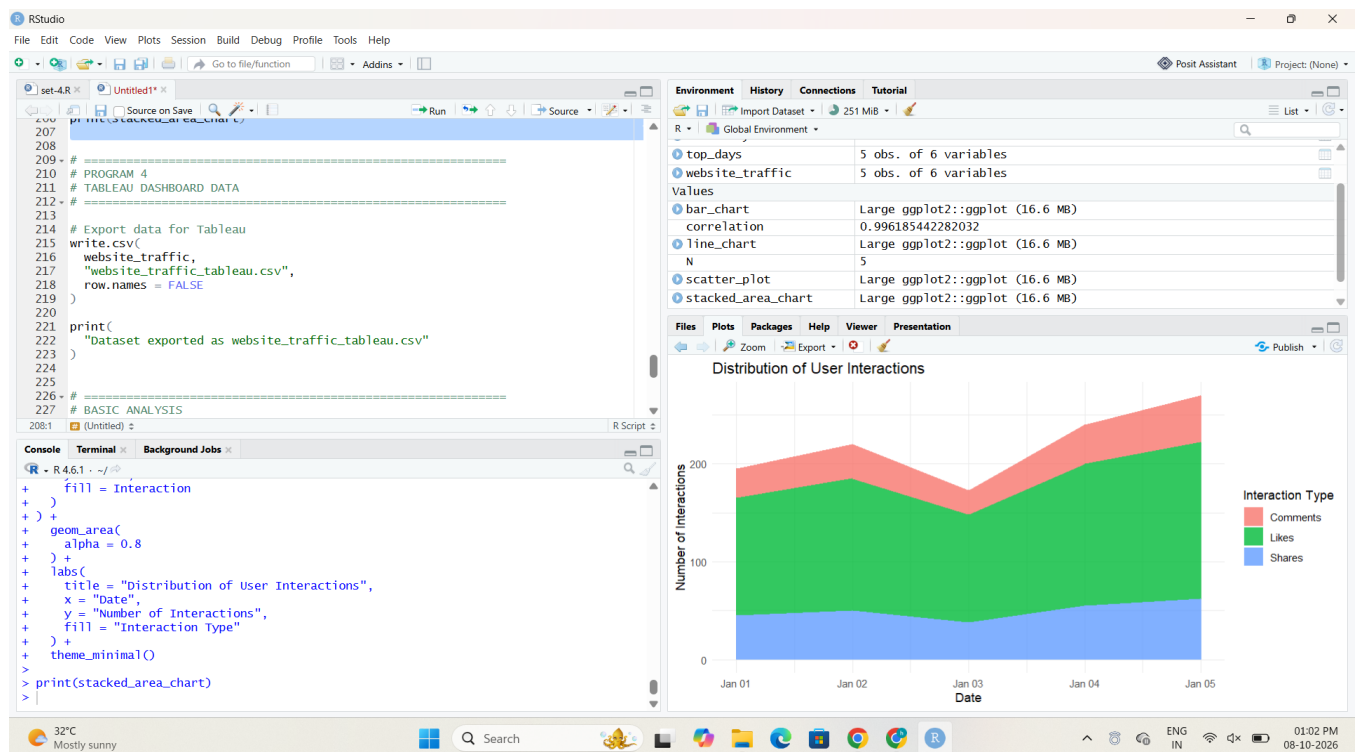


Figure 3: RStudio output – Distribution of User Interactions

Observation: Likes form the largest portion of interactions on each day, while shares and comments contribute smaller portions. The total sample interaction volume is highest on January 5.

Program 4: Tableau Dashboard – Website Traffic

The R program exports the website traffic data to website_traffic_tableau.csv. The dashboard below is an illustrative Tableau-style representation created from the same supplied data. It demonstrates the requested line chart and bar chart arrangement; it is not a screenshot captured from Tableau.

```
# PROGRAM 4: TABLEAU DASHBOARD DATA
# -----
write.csv(
  website_traffic,
  "website_traffic_tableau.csv",
  row.names = FALSE
)

print("Dataset exported as website_traffic_tableau.csv")

# -----
```

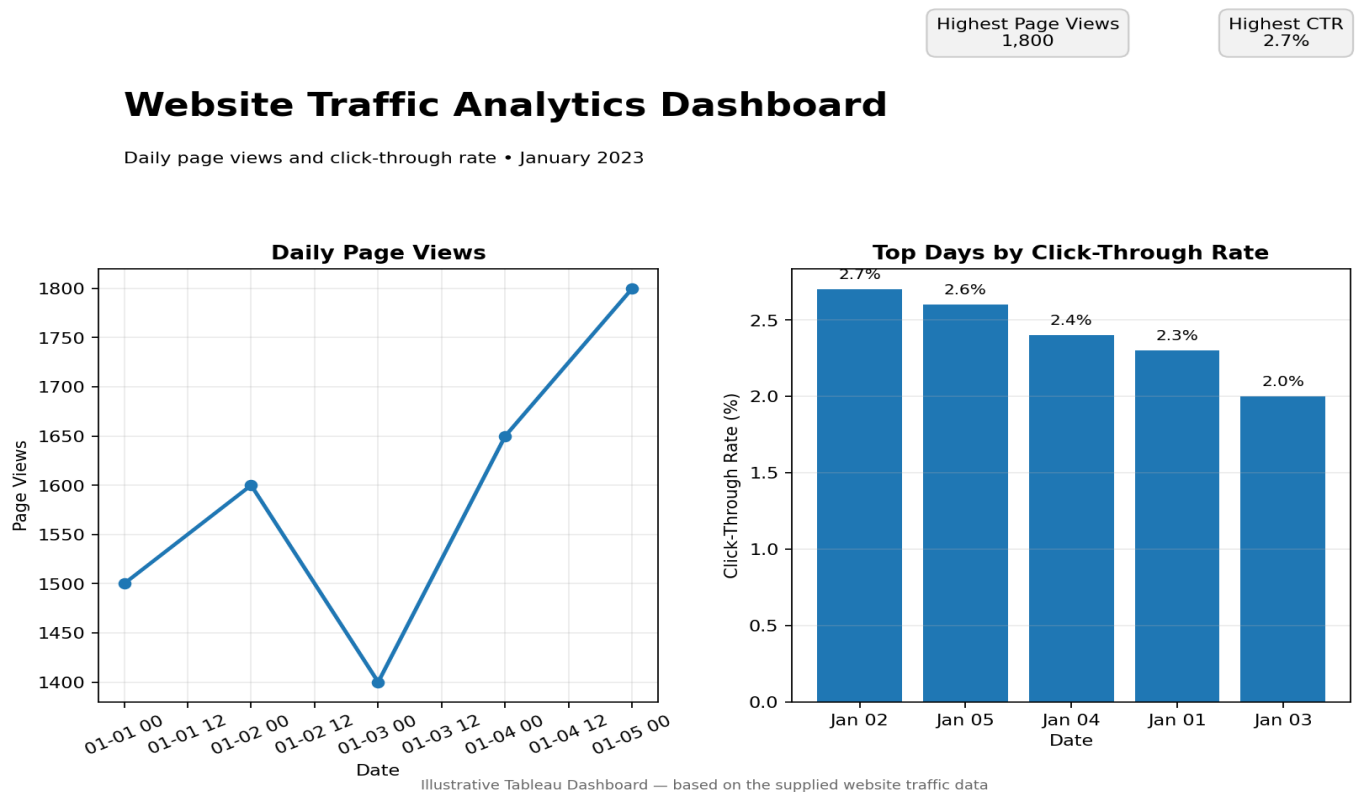


Figure 4: Illustrative Tableau-style dashboard – Website Traffic Analytics

Dashboard interpretation: The line chart highlights the daily page-view trend, while the bar chart ranks days by click-through rate. The highest page views occur on January 5, while the highest click-through rate occurs on January 2.

Overall Result

The Website Analytics experiment was completed successfully. The line chart visualized daily page views, the bar chart ranked the top days by click-through rate, the stacked area chart represented user interaction categories, and the website traffic data was exported for dashboard development. The analysis shows a final-day peak in page views and a January 2 peak in click-through rate.

Key Findings

- Highest page views: 1,800 on 2023-01-05.
- Lowest page views: 1,400 on 2023-01-03.
- Highest click-through rate: 2.7% on 2023-01-02.
- Lowest click-through rate: 2.0% on 2023-01-03.
- The dashboard combines traffic trend and engagement-rate views for easier comparison.

Reference

ggplot2 documentation was used as the technical reference for line, column/bar and area geoms.